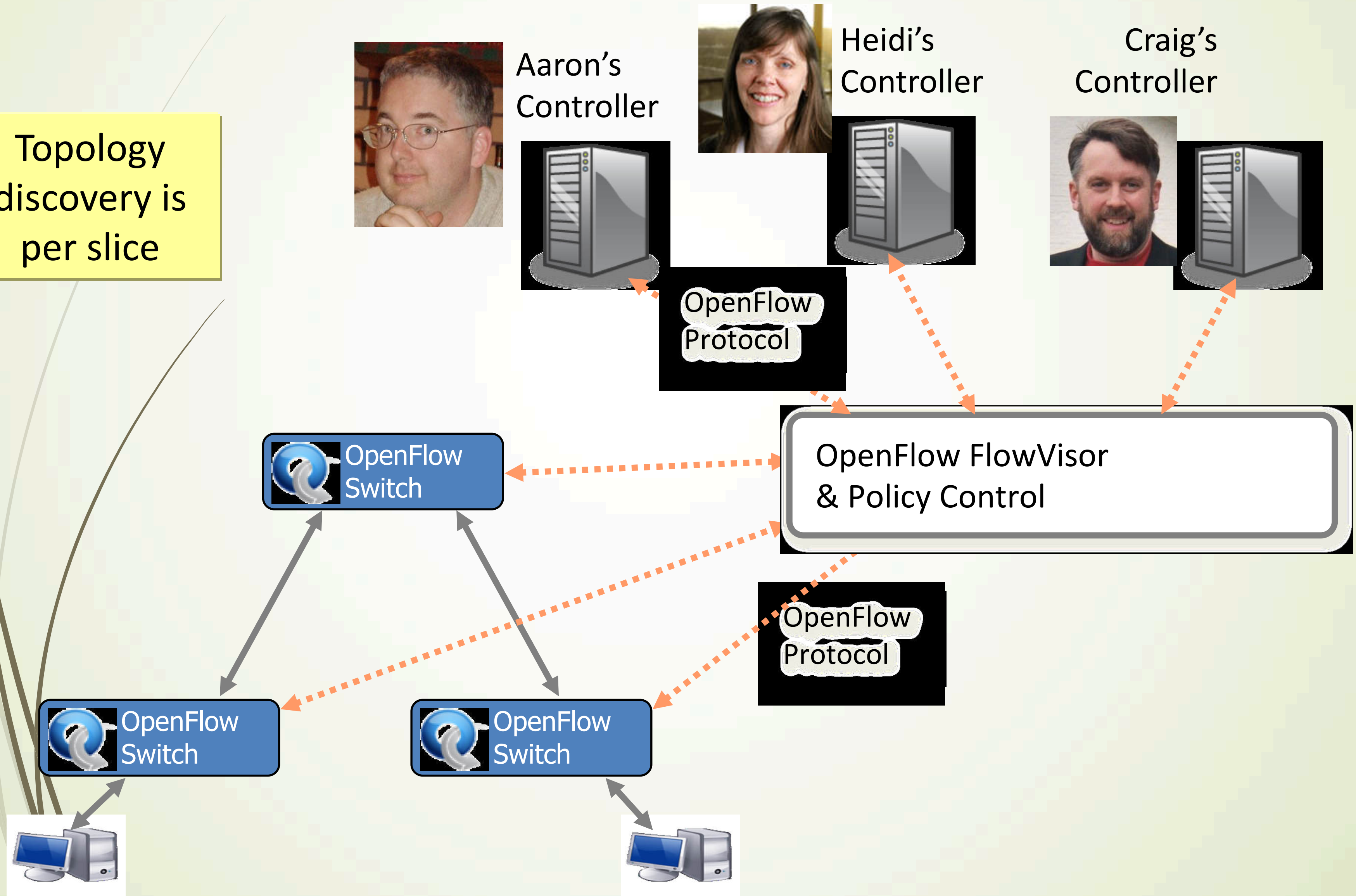
Agenda

- Motivations
- Concepts
- OpenFlow Protocol
- Network Virtualization
- Virtual Switches

FlowVisor for Network Virtualization

Topology discovery is per slice



FlowVisor Slicing

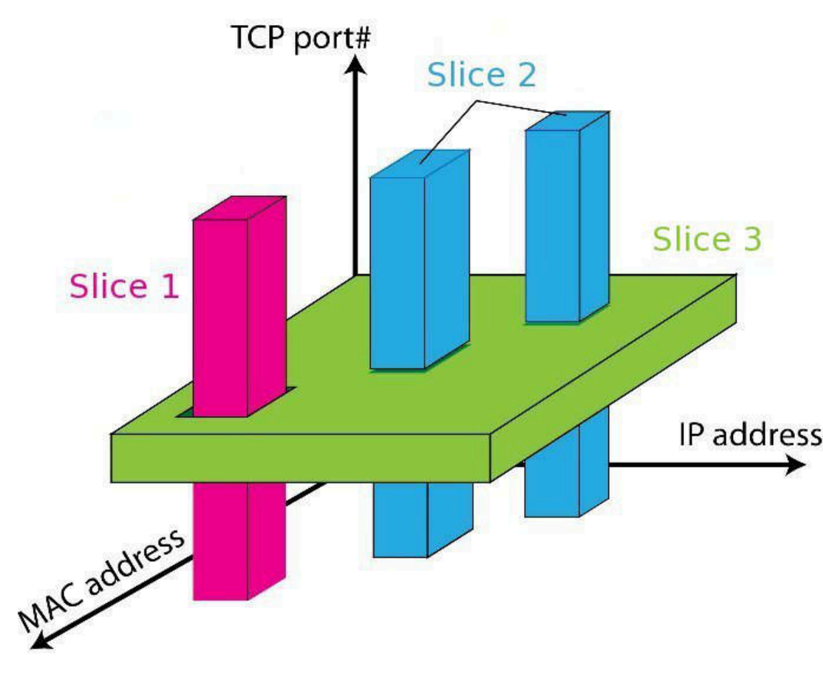
- Slices are defined using a slice definition policy
  - The policy language specifies the slice's resource limits, flowspace, and controller's location in terms of IP and TCP port-pair
  - FlowVisor enforces transparency and isolation between slices by inspecting, rewriting, and policing OpenFlow messages as they pass

Resource Limits Assigned in FlowVisor

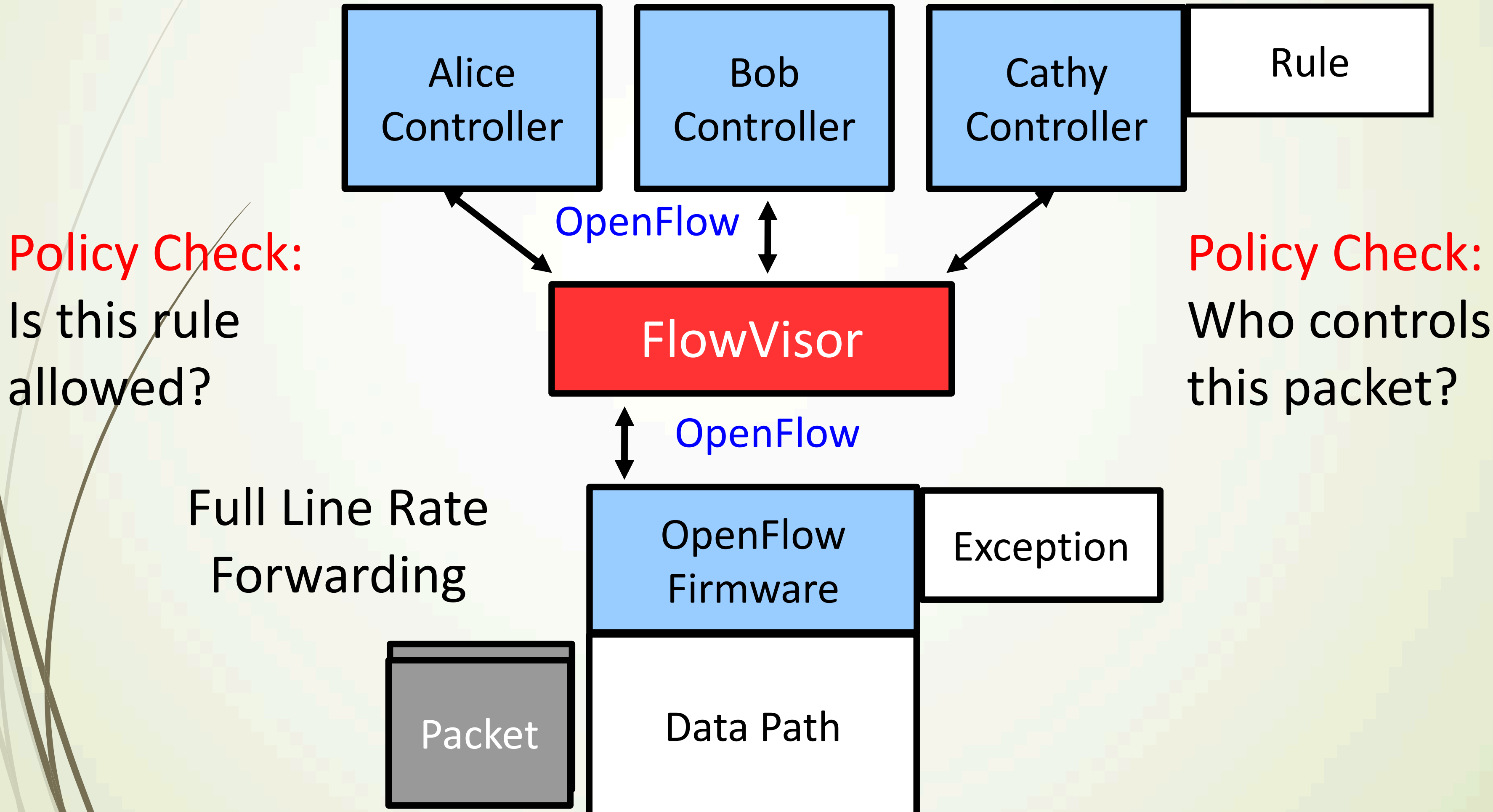
- Topology
  - Network devices
  - Physical ports
- Bandwidth
  - Each slice can be assigned a per port queue with a fraction of the total bandwidth
- CPU
  - Employs Course Rate Limiting techniques to keep new flow events from one slice from overrunning the CPU
- Forwarding Tables
  - Each slice has a finite quota of forwarding rules per device

FlowSpace Determine Assignments of Packets to Slices

- Source/Destination MAC address
- VLAN ID
- Ethertype
- IP protocol
- Source/Destination IP address
- ToS/DSCP
- Source/Destination port number



OpenFlow Messages are Intercepted by FlowVisor



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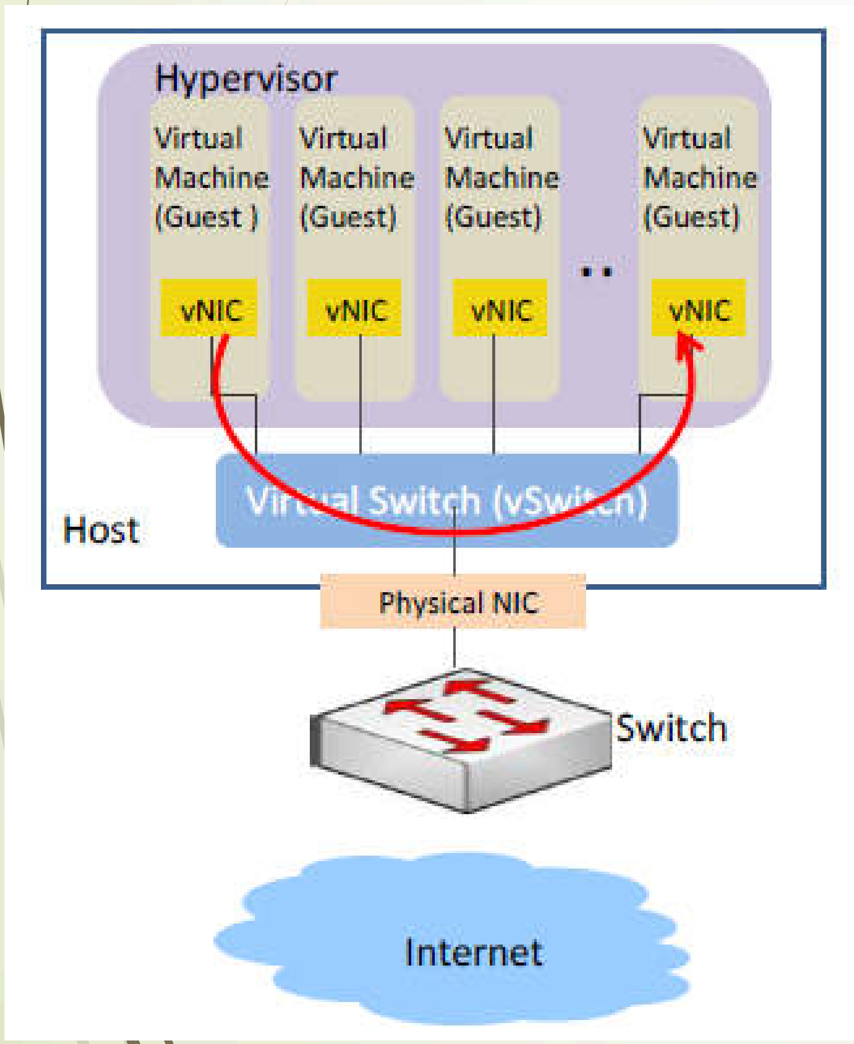
Virtual Switches

- Due to the cloud computing service, the number of virtual switches begins to expand dramatically
  - Management complexity
  - Security issues
  - Performance degradation
- Machine virtualization is enhanced by:
  - Software or hardware based virtual switches

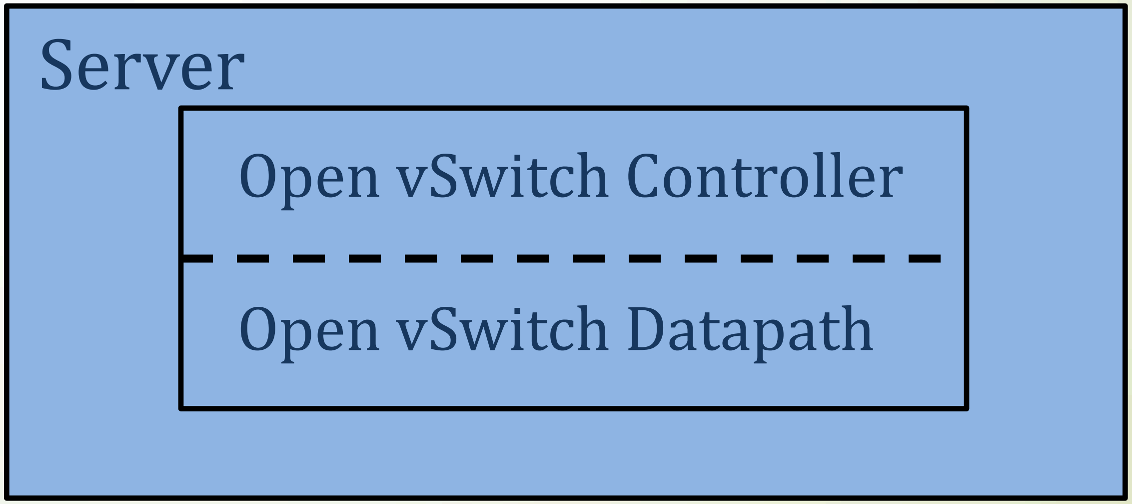
Limitations of vSwitches (before SDN)



- The hypervisors implement vSwitch
- Each VM has at least one virtual network interface cards (vNICs) and shared physical network interface cards (pNICs) on the physical host through vSwitch
- Administrators don't have effective solution to separate packets from different VM users
- For VMs reside in the same physical machine, their traffic visibility is a big issue

Open vSwitch

- A software-based solution
  - Resolve the problems of network separation and traffic visibility, so the cloud users can be assigned VMs with elastic and secure network configurations
- Flexible Controller in User-Space
- Fast Datapath in Kernel



Features of Open vSwitch

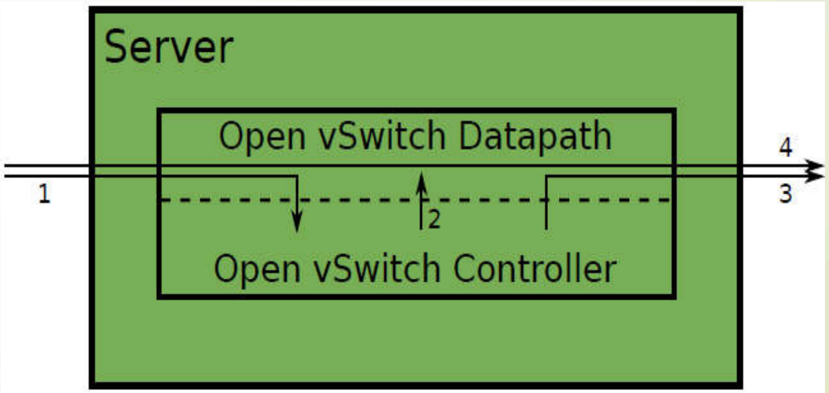
- Multiple ports to physical switches
  - A port may have one or more interfaces
    - Bonding allows more than once interface per port
- Centralized control through OpenFlow
- Works on Linux-based hypervisors:
  - Xen
  - KVM
  - VirtualBox
- IEEE 802.1Q Support
  - Enable virtual LAN function
  - By attaching VLAN ID to Linux virtual interfaces, each user will have its own LAN environment separated from other users

Open vSwitch Defines Flows as…

- Any combination of
  - Input port
  - VLAN ID (802.1Q)
  - Ethernet Source MAC address
  - Ethernet Destination MAC address
  - IP Source MAC address
  - IP Destination MAC address
  - TCP/UDP/... Source Port
  - TCP/UDP/... Destination Port

Open vSwitch Operates as Other OpenFlow Switches

- The 1st packet of a flow is sent to the controller
- The controller programs the datapath's actions for a flow
  - Usually one, but may be a list
  - Actions include:
    - Forward to a port or ports
    - mirror
    - Encapsulate and forward to controller
    - Drop
- And returns the packet to the datapath
- Subsequent packets are handled directly by the datapath



Questions

chsu@cs.nthu.edu.tw